Predicting phase acquisition between coupled PING circuits

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Abstract

This planned experiment asks whether reciprocal excitation from each PING circuit's excitatory population to the other circuit's inhibitory population can acquire gamma synchrony over several cycles. The protocol will first measure how one mature PING rhythm responds to an E-to-I pulse. It will use that response to predict the existence, phase lag, and convergence rate of a stable coupled state before running two circuits. The decisive trial must visibly separate an unsynchronized baseline, progressive acquisition, and a synchronized state.

Methods

1. **Calibrate two independent mature PING oscillators.** Construct two 800 E / 200 I circuits with independent Poisson drive and zero cross-circuit weight. Select fixed intrinsic and drive parameters that produce sustained gamma while retaining enough natural-frequency detuning to generate visible phase drift. Report E and I firing rates, natural frequency, cycle variability, and uncoupled phase drift over a small calibration seed set. Reject silent, saturated, intermittent, or already phase-stationary pairs, then freeze the parameters and reserve separate mature states and future-input seeds for validation.
2. **Measure the macroscopic E-to-I phase-response curve.** Give the graph a dedicated spike input connected to I using the same declared connection rule, weight, AMPA kinetics, and axonal delay intended for cross-circuit coupling. Replay E-volley spike patterns measured during Step 1, preserving their size and temporal dispersion rather than synchronously stimulating every I neuron. At evenly spaced phases of a mature E cycle, compare a sender-equivalent volley with an identical-state, identical-input silent continuation. Define the response as the phase displacement after the rhythm has recovered; retain the one-cycle displacement only as a transient diagnostic. Record changes in volley size, additional or merged I volleys, and suppressed or skipped E volleys. This establishes both the phase response and the amplitude-changing boundary beyond which a phase-only description is inadequate.
3. **Predict the coupled phase dynamics.** Combine the measured phase-response curve, sender-volley waveform, and candidate delays to estimate the interaction function for two reciprocally coupled rhythms. Restrict prediction to strengths that primarily shift phase and preserve the PING cycle. Predeclare one condition predicted to lock, one predicted not to lock, the stable lag, the common mean frequency, and a convergence-time range. Extra volleys, skipped cycles, suppression, or a regime transition are phase-model violations rather than failed simulations.

4. **Test predicted synchronization acquisition.** Compile structurally identical zero-weight and coupled graphs containing both reciprocal E-to-I projections. Run the zero-weight graph through burn-in and at least 500 ms of recorded pre-switch activity, then continue its complete runtime state in the coupled graph using the next segment of the same pre-generated inputs. Exact split-run parity and first-effect timing are engineering acceptance tests for this protocol. The two executor calls must form one causal trajectory; no spike train may be shifted or selected using its post-switch outcome.

   Run both predeclared conditions for at least three predicted convergence times after the switch, with a minimum post-switch duration of 2000 ms. Use E-population volley times as the primary phase diagnostic. Plot sampled rasters for circuits A and B, separate A and B E-rate panels on a common grid, and cycle-by-cycle A-minus-B phase. Report mean-frequency convergence as a summary statistic rather than differentiating the noisy phase trace into another primary plot. A successful locking trajectory must show pre-switch drift, progressive phase correction, and preregistered residence around the predicted lag. Compare observed lag and convergence time with the Step 3 prediction and report later slips or amplitude disruptions without cropping them.

5. **Replicate the prediction test.** Repeat the locking and non-locking conditions across the reserved mature states and future-input seeds. Apply the same analysis to every run. Report acquisition probability, convergence-time distribution, phase-lag error, post-acquisition residence, phase slips, mean frequency difference, and population health. A representative raster may illustrate the mechanism, but the verdict comes from the complete held-out set. Reciprocal E-to-I coupling explains acquisition only if the sender-equivalent phase response predicts the observed lag, stability, common frequency, and convergence timescale without materially changing cycle amplitude. Treat E-to-I as an isolated effective motif; E-to-E and mixed coupling are future experiments.

Implementation boundary

The first implementation requires no general time-dependent-weight feature. *snnlang* can already declare independent spike inputs, fixed AMPA and GABA projections, reciprocal feedback paths, delays, and the two otherwise identical zero-weight and coupled graphs. *tools/snn* can already continue complete runtime state across those graphs because its compatibility signature excludes parameter values while retaining structural checks. The runner must own input generation, phase-target selection, branching, response-curve estimation, phase-model fitting, condition grids, and every plot.

The present system cannot change a weight or trigger an intervention from an online phase estimate within one executor call. It also has no arbitrary analog conductance-input primitive. Those capabilities are not prerequisites here: a declared spike volley through a fixed AMPA projection supplies the phase probe, and exact split-run continuation supplies the coupling switch. Add a schedule or event-controller abstraction only if a later experiment genuinely requires continuously varying parameters or closed-loop intervention.

Results

This experiment is planned and has no results yet.